LAB 2

TELEMATIC SERVICES ON INTERNET

Subgrupo 1
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Domain #1: peralesgomez

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Domain #2: garciazafra

BACKGROUND

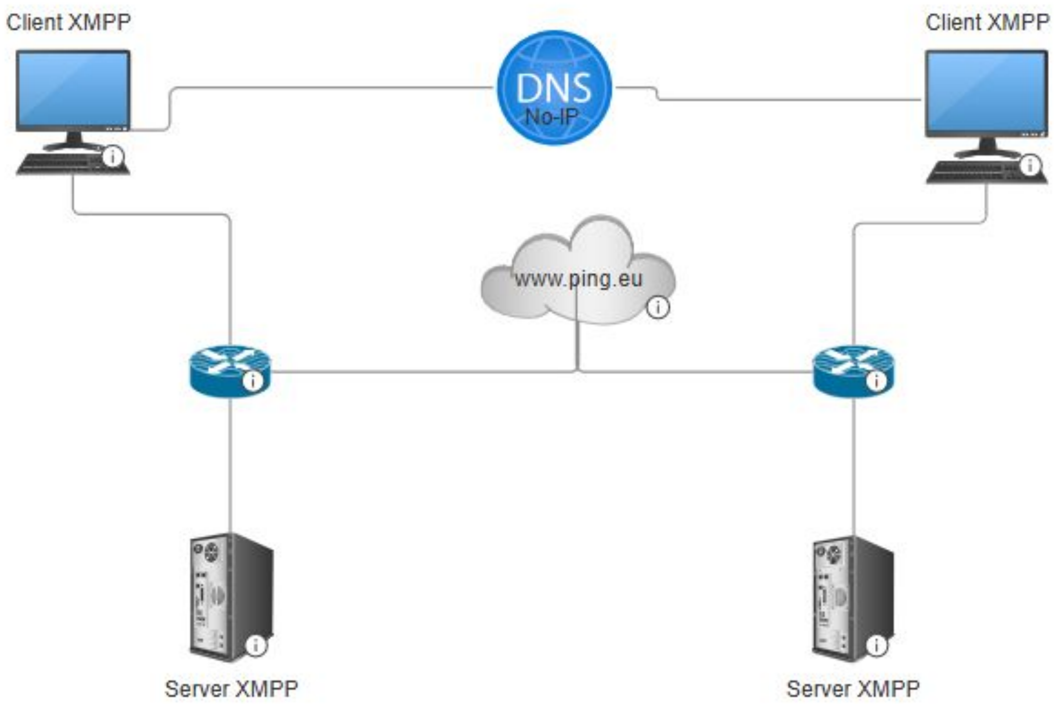
En esta práctica se ha realizado al despliegue de servidores XMPP en Ubuntu y Windows usando Docker, basados en ejabberd, en dos redes corporativas remotas accesibles desde Internet utilizando conexiones ADSL domésticas.

Además, debido a la ausencia de direcciones ip públicas fijas se ha empleado el servicio DDNS para la accesibilidad desde el exterior. Se han creado dominios diferenciados para cada subequipo utilizando nombres personalizados a través de no-ip.com. También se ha empleado Pidgin como cliente de mensajería.

Adicionalmente, se ha implementado un segundo servicio consistente en un despliegue de un servidor Jellyfin en Ubuntu, con el objetivo de permitir la transmisión y visualización de contenidos multimedia accesibles vía web.

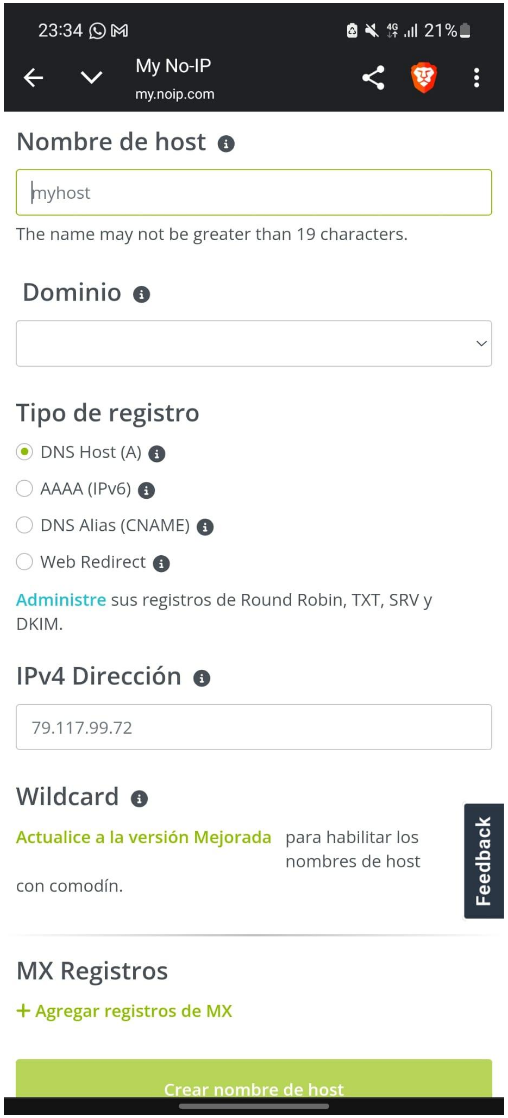
Durante el desarrollo se han utilizado herramientas de diagnóstico como ping, traceroute, netstat y wireshark.
La metodología seguida ha sido estructurada por fases: planificación, configuración de servicios, pruebas entre dominios, resolución de errores y validación final.

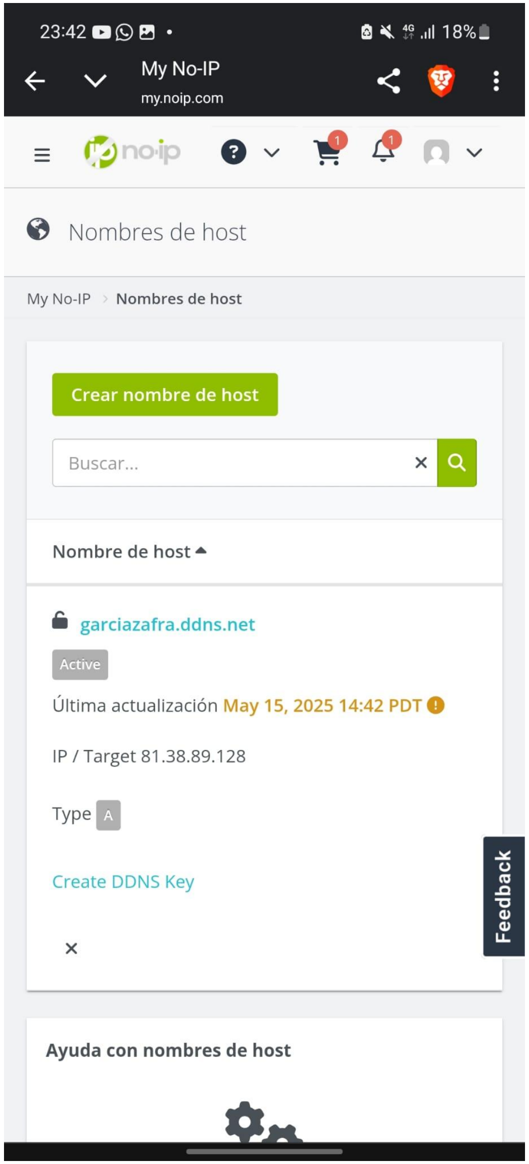
BACKGROUND


Client XMPP
DNS
No-IP
Client XMPP
www.ping.eu
Server XMPP
Server XMPP

CREATE Dynamic DNS Account: domain1

CREATE Dynamic DNS Account: domain2





CHECK domain1 with nslookup and wireshark

The Cloudflare DNS server responded with these DNS records. Cloudflare will serve these records for as long as the time to live (TTL) has not expired. After this period, Cloudflare will update its cache by querying one of the authoritative name servers.

A records

IPv4 address	Revalidate in
86.127.242.23	1m

AAAA records

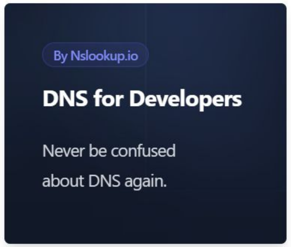
No AAAA records found.

CNAME record

No CNAME record found.

TXT records

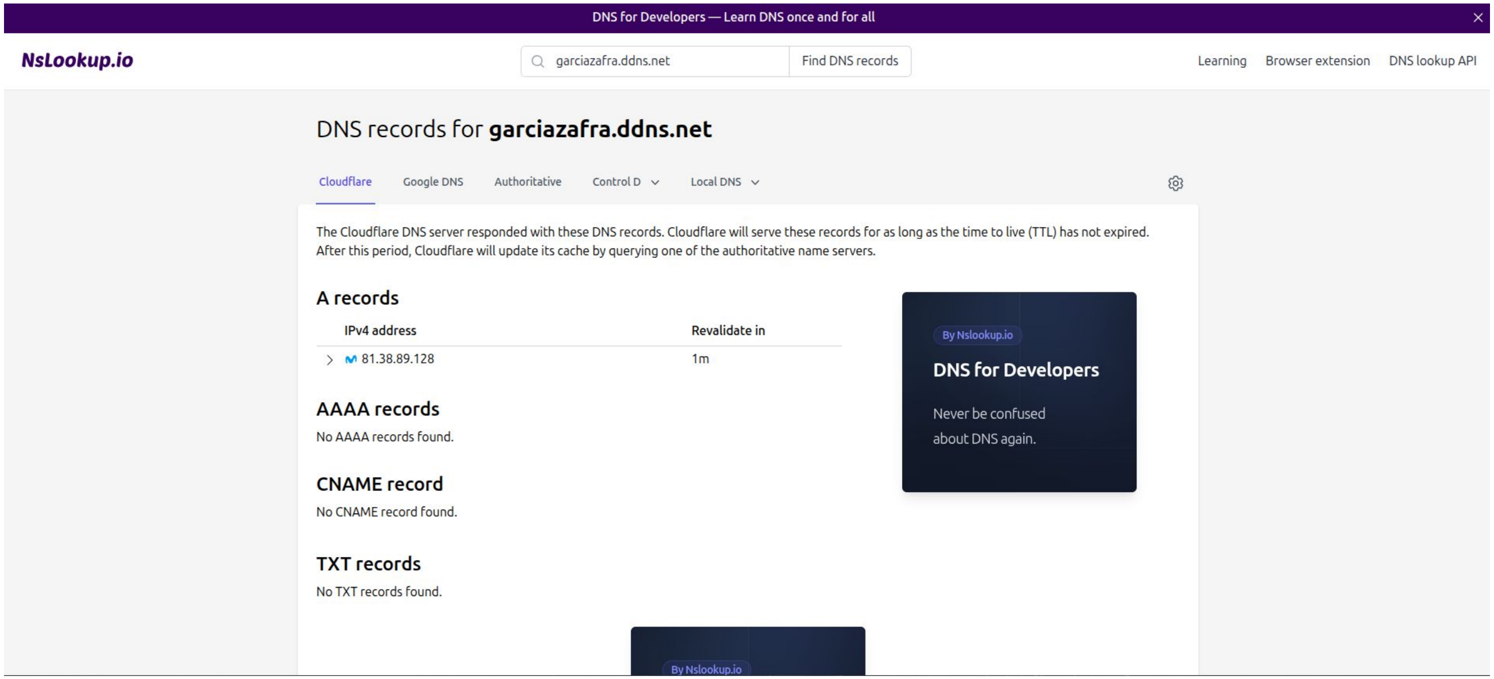
No TXT records found.



dns

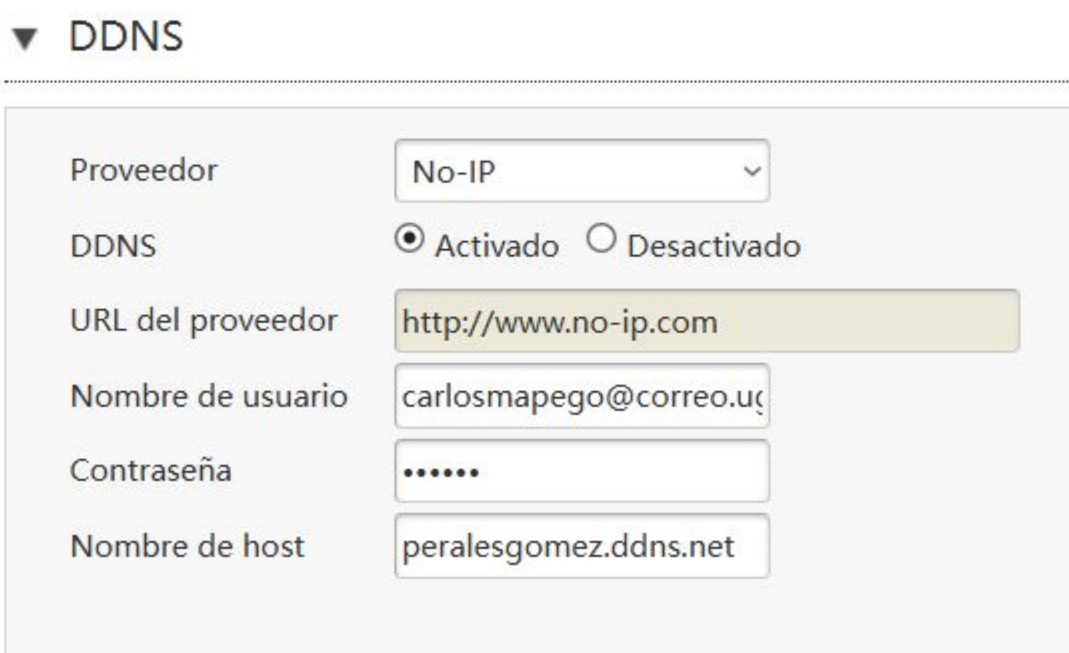
No.	Time	Source	Destination	Protocol	Length	Info
157	11.734272	2a0c:5a83:920b:7700…	2a0c:5a80:0:2::1	DNS	101	Standard query 0xe32c AAAA peralesgomez.ddns.net
158	11.757627	2a0c:5a83:920b:7700…	2a0c:5a84:0:2::1	DNS	101	Standard query 0xe32c AAAA peralesgomez.ddns.net
159	11.767342	2a0c:5a80:0:2::1	2a0c:5a83:920b:7700…	DNS	161	Standard query response 0xe32c AAAA peralesgomez.ddns.net SOA nf1.no-ip.com
161	11.788239	2a0c:5a84:0:2::1	2a0c:5a83:920b:7700…	DNS	161	Standard query response 0xe32c AAAA peralesgomez.ddns.net SOA nf1.no-ip.com

CHECK domain2 with nslookup and wireshark




```
1930 136.413469   192.168.1.135       100.90.1.1          DNS     73 Standard query 0x8a29 A p13n.adobe.io
1938 138.148168   2a0c:5a83:920b:7700… 2a0c:5a80:0:2::1    DNS    100 Standard query 0x8acb A garciazafra.ddns.net
1939 138.148464   2a0c:5a83:920b:7700… 2a0c:5a80:0:2::1    DNS    100 Standard query 0x383d AAAA garciazafra.ddns.net
1940 138.161595   2a0c:5a80:0:2::1    2a0c:5a83:920b:7700… DNS    116 Standard query response 0x8acb A garciazafra.ddns.net A 81.38.89.128
1941 138.174913   2a0c:5a83:920b:7700… 2a0c:5a84:0:2::1    DNS    100 Standard query 0x383d AAAA garciazafra.ddns.net
```

Configure DDNS at home domain1



Configure DDNS at home domain2

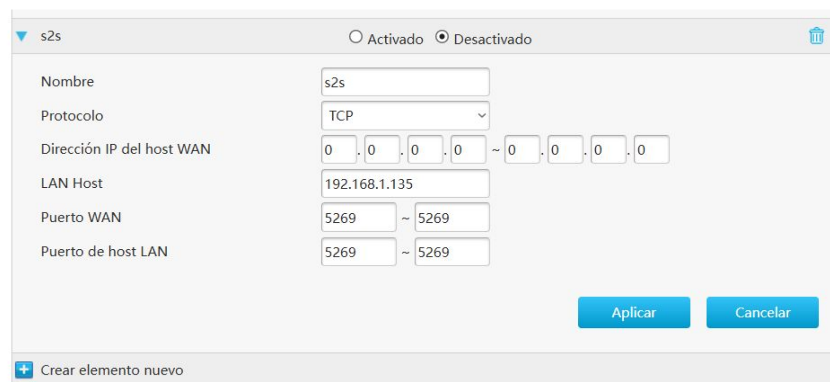
Dynamic DNS

The Dynamic DNS service allows you to alias a dynamic IP address to a static hostname in any of the many domains, allowing your Broadband Router to be more easily accessed from various locations on the Internet.

Choose Add or Remove to configure Dynamic DNS.

Hostname	Username	Service	Interface	Remove
garciazafra.ddns.net	garciazafra	noip	veip0.2	☐

Open ports in ADSL router domain1

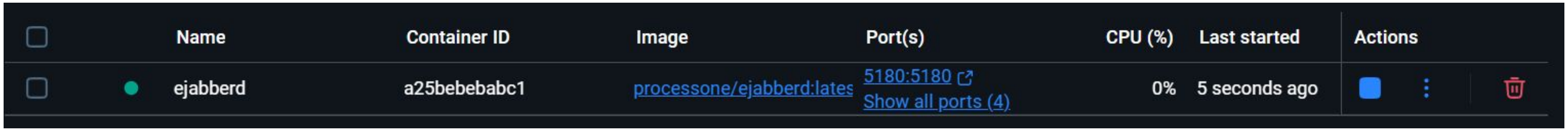


Open ports in ADSL router domain2

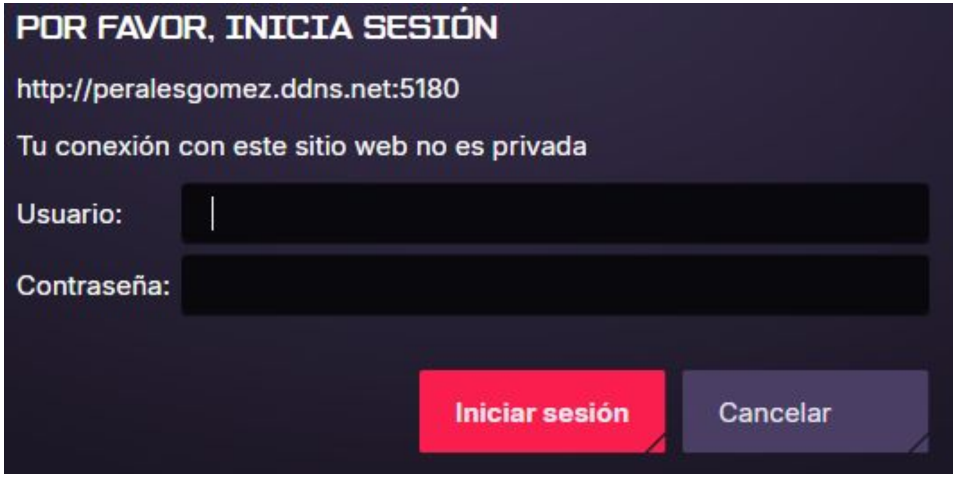
Editar

Nombre	Protocolo	Puerto/Rango Externo	Puerto/Rango Interno	Direccion IP	Activar
c2s	TCP	5222:5222	5222:5222	192.168.1.93	ON
s2s	TCP	5269:5269	5269:5269	192.168.1.93	ON
Jellyfin	TCP	8096:8096	8096:8096	192.168.1.93	ON

Install service 1




```
2025-05-18 10:52:40.637 [info] Start accepting UDP connections at [::]:5478 for :ejabberd_stun
2025-05-18 10:52:40.637 [info] Start accepting TCP connections at [::]:1883 for :mod_mqtt
2025-05-18 10:52:40.637 [info] Start accepting TCP connections at [::]:5222 for :ejabberd_c2s
2025-05-18 10:52:40.637 [info] Start accepting TCP connections at [::]:5269 for :ejabberd_s2s_in
2025-05-18 10:52:40.637 [info] Start accepting TCP connections at [::]:1880 for :ejabberd_http
2025-05-18 10:52:40.637 [info] Start accepting TCP connections at 127.0.0.1:7777 for :mod_proxy65_stream
2025-05-18 10:52:40.637 [info] Start accepting TLS connections at [::]:5223 for :ejabberd_c2s
2025-05-18 10:52:40.637 [info] Start accepting TCP connections at [::]:5180 for :ejabberd_http
```




```
~ $ ejabberdctl register admin peralesgomez.ddns.net admin
User admin@peralesgomez.ddns.net successfully registered
~ $ ejabberdctl register garciazafra peralesgomez.ddns.net tstc_1234!
User garciazafra@peralesgomez.ddns.net successfully registered
~ $ ejabberdctl register profesor peralesgomez.ddns.net tstc_1234!
User profesor@peralesgomez.ddns.net successfully registered
~ $
```

Install service 1

```
PS C:\Users\usuario\Desktop> openssl req -x509 -nodes -days 3650 -newkey rsa:4096 -keyout privkey.pem -out fullchain.pem
 -config cert.conf -extensions v3_req
```

```
## If you already have certificates, list them here
certfiles:
  - /opt/ejabberd/conf/fullchain.pem
  - /opt/ejabberd/conf/privkey.pem
```

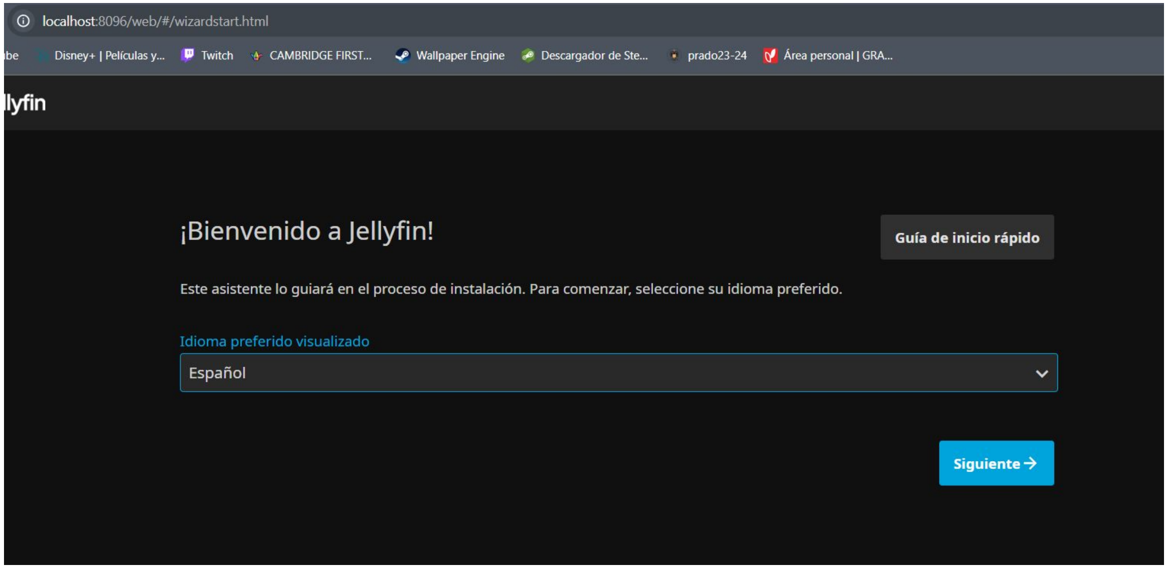
```
hosts:
  - peralesgomez.ddns.net
```

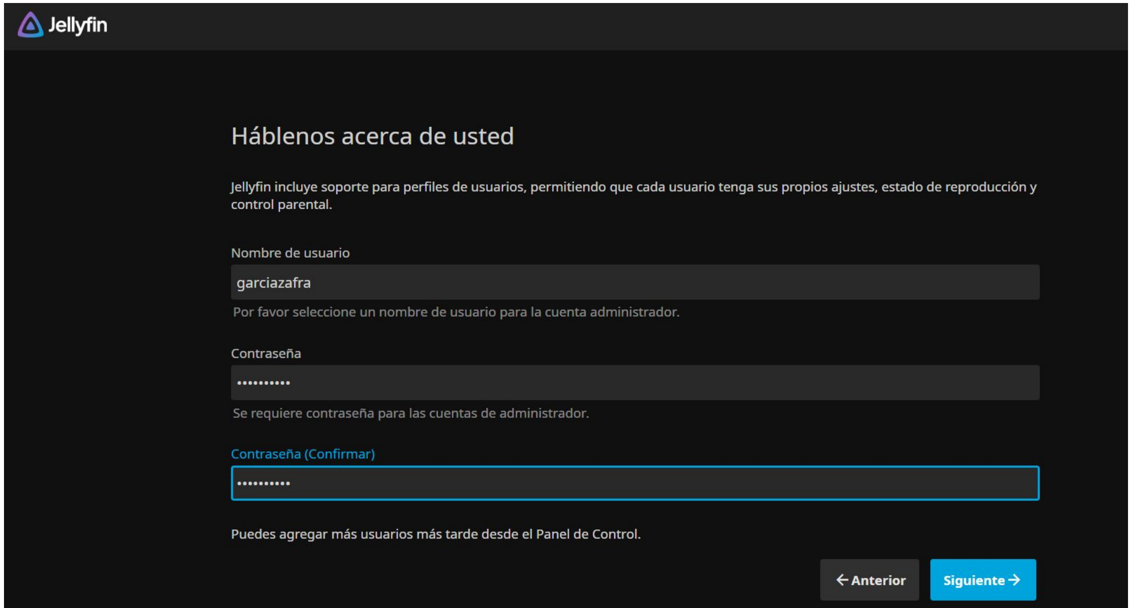
Install service 2

```
sudo apt purge jellyfin jellyfin-server jellyfin-web jellyfin-ffmpeg7
sudo apt autoremove --purge
sudo rm -rf /var/lib/jellyfin /etc/jellyfin /usr/lib/jellyfin /var/log/jellyfin
sudo apt install apt-transport-https gnupg curl
curl -fsSL https://repo.jellyfin.org/jellyfin_team.gpg.key | sudo gpg --dearmor -o /usr/share/keyrings/jellyfin-archive-keyring.gpg
echo "deb [signed-by=/usr/share/keyrings/jellyfin-archive-keyring.gpg] https://repo.jellyfin.org/ubuntu jammy main" | sudo tee /etc/apt/sources.list.d/jellyfin.list
sudo apt update
sudo apt install jellyfin
sudo systemctl start jellyfin
```

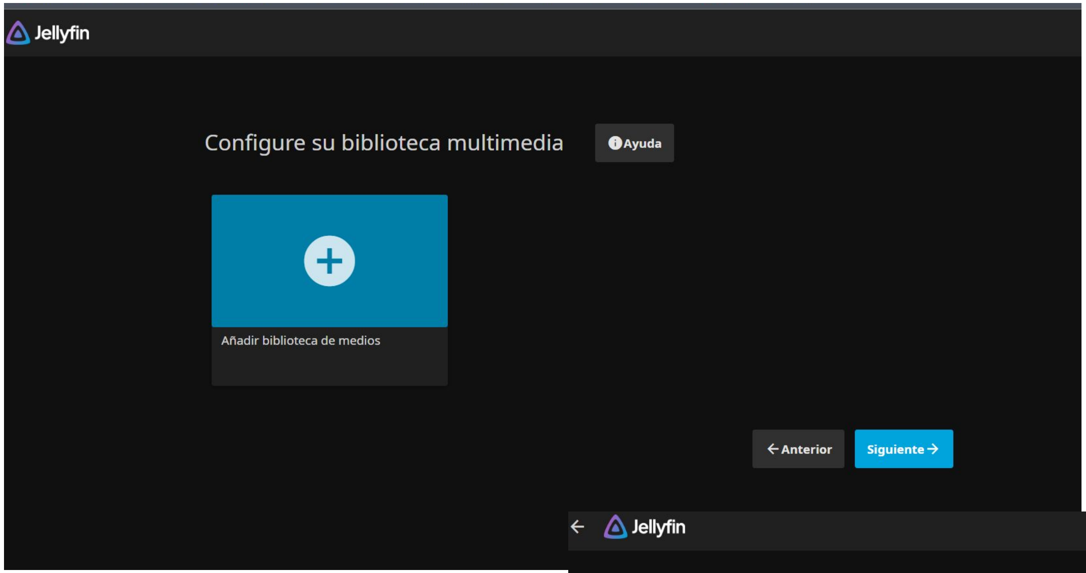
```
● jellyfin.service - Jellyfin Media Server
     Loaded: loaded (/usr/lib/systemd/system/jellyfin.service; enabled; preset: enabled)
    Drop-In: /etc/systemd/system/jellyfin.service.d
             └─jellyfin.service.conf
     Active: active (running) since Sun 2025-05-18 19:42:47 CEST; 1h 7min ago
   Main PID: 2486 (jellyfin)
      Tasks: 17 (limit: 18626)
     Memory: 345.8M (peak: 355.7M)
        CPU: 23.921s
     CGroup: /system.slice/jellyfin.service
             └─2486 /usr/bin/jellyfin --webdir=/usr/share/jellyfin/web --ffmpeg=/usr/lib/jellyfin-ffmpeg/ffmpeg
```

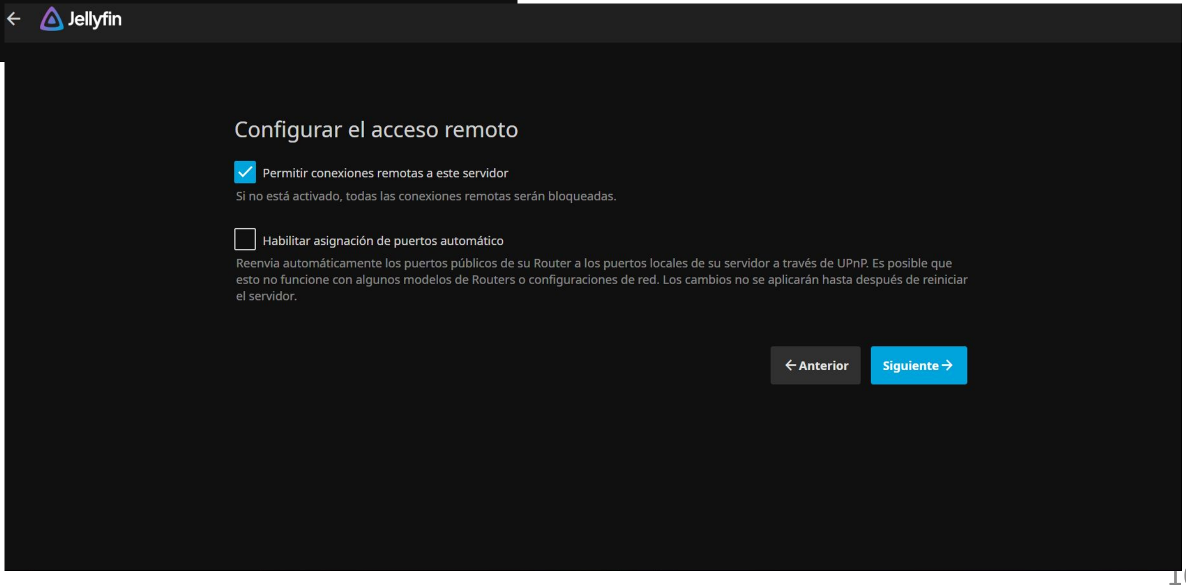
Install service 2





Install service 2





Check ports domain1 in your server with netstat -a

```
TCP    0.0.0.0:5222           DESKTOP-MLPRRIK:0      LISTENING
TCP    0.0.0.0:5269           DESKTOP-MLPRRIK:0      LISTENING
```

Check ports domain2 in your server with netstat -a

```
tcp6       0      0 :::5269                 :::*                    ESCUCHAR
tcp6       0      0 :::5223                 :::*                    ESCUCHAR
tcp6       0      0 :::5222                 :::*                    ESCUCHAR
tcp        0      0 0.0.0.0:8096            0.0.0.0:*               ESCUCHAR
```

Check open ports domain1 from outside with www.ping.eu

tcp.port == 5222

No.	Time	Source	Destination	Protocol	Length	Info
35	0.240135	192.168.1.135	86.127.247.32	TCP	77	11790 → 5222 [PSH, ACK] Seq=1 Ack=1 Win=254 Len=23 [TCP segment of a reassembled PDU]
36	0.241163	86.127.247.32	192.168.1.135	TCP	77	11790 → 5222 [PSH, ACK] Seq=1 Ack=1 Win=254 Len=23 [TCP segment of a reassembled PDU]
44	0.292566	192.168.1.135	86.127.247.32	TCP	54	5222 → 11790 [ACK] Seq=1 Ack=24 Win=251 Len=0
45	0.293856	86.127.247.32	192.168.1.135	TCP	54	5222 → 11790 [ACK] Seq=1 Ack=24 Win=251 Len=0
260	2.204166	88.198.46.51	192.168.1.135	TCP	74	58114 → 5222 [SYN] Seq=0 Win=7300 Len=0 MSS=1452 SACK_PERM TSval=1705897959 TSecr=0 WS=256
261	2.204467	192.168.1.135	88.198.46.51	TCP	74	5222 → 58114 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=186334970 TSecr=1705897
271	2.257295	88.198.46.51	192.168.1.135	TCP	66	58114 → 5222 [FIN, ACK] Seq=1 Ack=1 Win=7424 Len=0 TSval=1705898011 TSecr=186334970
272	2.257295	88.198.46.51	192.168.1.135	TCP	66	[TCP Keep-Alive] 58114 → 5222 [ACK] Seq=1 Ack=1 Win=7424 Len=0 TSval=1705898011 TSecr=186334970
275	2.257551	192.168.1.135	88.198.46.51	TCP	66	5222 → 58114 [ACK] Seq=1 Ack=2 Win=65280 Len=0 TSval=186335024 TSecr=1705898011
277	2.262824	192.168.1.135	88.198.46.51	TCP	66	5222 → 58114 [FIN, ACK] Seq=1 Ack=2 Win=65280 Len=0 TSval=186335029 TSecr=1705898011
282	2.316279	88.198.46.51	192.168.1.135	TCP	66	58114 → 5222 [ACK] Seq=2 Ack=2 Win=7424 Len=0 TSval=1705898069 TSecr=186335029
399	3.244956	192.168.1.135	86.127.247.32	TCP	77	11784 → 5222 [PSH, ACK] Seq=1 Ack=1 Win=255 Len=23 [TCP segment of a reassembled PDU]
400	3.246568	86.127.247.32	192.168.1.135	TCP	77	11784 → 5222 [PSH, ACK] Seq=1 Ack=1 Win=255 Len=23 [TCP segment of a reassembled PDU]
410	3.299068	192.168.1.135	86.127.247.32	TCP	54	5222 → 11784 [ACK] Seq=1 Ack=24 Win=255 Len=0
412	3.300107	86.127.247.32	192.168.1.135	TCP	54	5222 → 11784 [ACK] Seq=1 Ack=24 Win=255 Len=0
953	5.238217	192.168.1.135	86.127.247.32	TCP	77	11790 → 5222 [PSH, ACK] Seq=24 Ack=1 Win=254 Len=23 [TCP segment of a reassembled PDU]
954	5.239811	86.127.247.32	192.168.1.135	TCP	77	11790 → 5222 [PSH, ACK] Seq=24 Ack=1 Win=254 Len=23 [TCP segment of a reassembled PDU]
961	5.286701	192.168.1.135	86.127.247.32	TCP	54	5222 → 11790 [ACK] Seq=1 Ack=47 Win=251 Len=0
963	5.288192	86.127.247.32	192.168.1.135	TCP	54	5222 → 11790 [ACK] Seq=1 Ack=47 Win=251 Len=0

Check open ports domain1 from outside with www.ping.eu

Your IP is 79.117.251.102

Online service Port check

Port check – Tests if TCP port is opened on specified IP

IP address or host name: 86.127.247.32 Port number: 5269 Enter code: Go

86.127.247.32:**5269** port is **open**

tcp.port == 5269

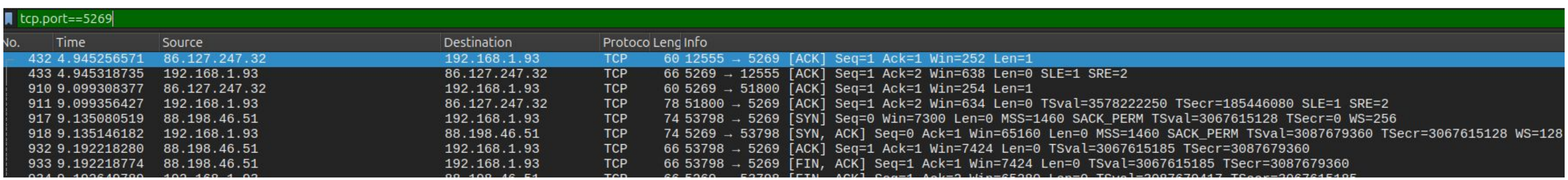
No.	Time	Source	Destination	Protocol	Length	Info
291	2.281797	192.168.1.135	86.127.247.32	TCP	77	11784 → 5222 [PSH, ACK] Seq=1 Ack=1 Win=254 Len=23 [TCP segment of a reassembled PDU]
292	2.283075	86.127.247.32	192.168.1.135	TCP	77	11784 → 5222 [PSH, ACK] Seq=1 Ack=1 Win=254 Len=23 [TCP segment of a reassembled PDU]
300	2.334910	192.168.1.135	86.127.247.32	TCP	54	5222 → 11784 [ACK] Seq=1 Ack=24 Win=511 Len=0
301	2.336980	86.127.247.32	192.168.1.135	TCP	54	5222 → 11784 [ACK] Seq=1 Ack=24 Win=511 Len=0
516	4.285377	192.168.1.135	86.127.247.32	TCP	77	11790 → 5222 [PSH, ACK] Seq=1 Ack=1 Win=251 Len=23 [TCP segment of a reassembled PDU]
517	4.286621	86.127.247.32	192.168.1.135	TCP	77	11790 → 5222 [PSH, ACK] Seq=1 Ack=1 Win=251 Len=23 [TCP segment of a reassembled PDU]
522	4.329222	192.168.1.135	86.127.247.32	TCP	54	5222 → 11790 [ACK] Seq=1 Ack=24 Win=252 Len=0
524	4.331262	86.127.247.32	192.168.1.135	TCP	54	5222 → 11790 [ACK] Seq=1 Ack=24 Win=252 Len=0
961	7.284722	192.168.1.135	86.127.247.32	TCP	77	11784 → 5222 [PSH, ACK] Seq=24 Ack=1 Win=254 Len=23 [TCP segment of a reassembled PDU]
962	7.285825	86.127.247.32	192.168.1.135	TCP	77	11784 → 5222 [PSH, ACK] Seq=24 Ack=1 Win=254 Len=23 [TCP segment of a reassembled PDU]
974	7.327009	192.168.1.135	86.127.247.32	TCP	54	5222 → 11784 [ACK] Seq=1 Ack=47 Win=511 Len=0
975	7.329486	86.127.247.32	192.168.1.135	TCP	54	5222 → 11784 [ACK] Seq=1 Ack=47 Win=511 Len=0
1256	9.286303	192.168.1.135	86.127.247.32	TCP	77	11790 → 5222 [PSH, ACK] Seq=24 Ack=1 Win=251 Len=23 [TCP segment of a reassembled PDU]
1257	9.287710	86.127.247.32	192.168.1.135	TCP	77	11790 → 5222 [PSH, ACK] Seq=24 Ack=1 Win=251 Len=23 [TCP segment of a reassembled PDU]
1282	9.330394	192.168.1.135	86.127.247.32	TCP	54	5222 → 11790 [ACK] Seq=1 Ack=47 Win=252 Len=0
1283	9.331476	86.127.247.32	192.168.1.135	TCP	54	5222 → 11790 [ACK] Seq=1 Ack=47 Win=252 Len=0

Check ports domain2 from outside with www.ping.eu

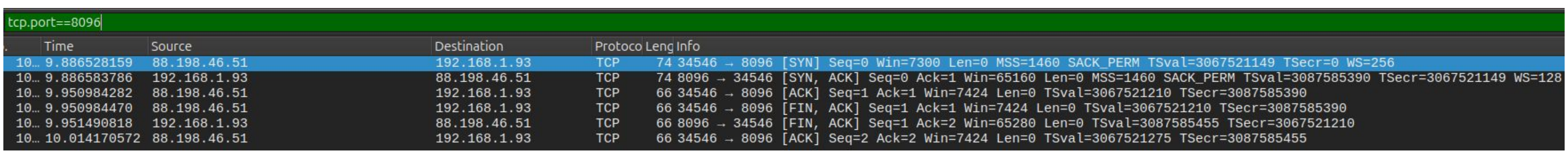
tcp.port==5222

No.	Time	Source	Destination	Protocol	Length	Info
500	3.664987757	88.198.46.51	192.168.1.93	TCP	74	38976 → 5222 [SYN] Seq=0 Win=7300 Len=0 MSS=1460 SACK_PERM TSval=3067344860 TSecr=0 WS=256
501	3.665034991	192.168.1.93	88.198.46.51	TCP	74	5222 → 38976 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3087409107 TSecr=3067344860 WS=128
511	3.723067592	88.198.46.51	192.168.1.93	TCP	66	38976 → 5222 [ACK] Seq=1 Ack=1 Win=7424 Len=0 TSval=3067344918 TSecr=3087409107
512	3.723067783	88.198.46.51	192.168.1.93	TCP	66	38976 → 5222 [FIN, ACK] Seq=1 Ack=1 Win=7424 Len=0 TSval=3067344918 TSecr=3087409107
513	3.723732374	192.168.1.93	88.198.46.51	TCP	66	5222 → 38976 [FIN, ACK] Seq=1 Ack=2 Win=65280 Len=0 TSval=3087409165 TSecr=3067344918
523	3.781983147	88.198.46.51	192.168.1.93	TCP	66	38976 → 5222 [ACK] Seq=2 Ack=2 Win=7424 Len=0 TSval=3067344977 TSecr=3087409165

Check ports domain2 from outside with www.ping.eu



Check ports domain2 from outside with www.ping.eu



tcp.port==8096

.	Time	Source	Destination	Protoco	Leng	Info
10…	9.886528159	88.198.46.51	192.168.1.93	TCP	74	34546 → 8096 [SYN] Seq=0 Win=7300 Len=0 MSS=1460 SACK_PERM TSval=3067521149 TSecr=0 WS=256
10…	9.886583786	192.168.1.93	88.198.46.51	TCP	74	8096 → 34546 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3087585390 TSecr=3067521149 WS=128
10…	9.950984282	88.198.46.51	192.168.1.93	TCP	66	34546 → 8096 [ACK] Seq=1 Ack=1 Win=7424 Len=0 TSval=3067521210 TSecr=3087585390
10…	9.950984470	88.198.46.51	192.168.1.93	TCP	66	34546 → 8096 [FIN, ACK] Seq=1 Ack=1 Win=7424 Len=0 TSval=3067521210 TSecr=3087585390
10…	9.951490818	192.168.1.93	88.198.46.51	TCP	66	8096 → 34546 [FIN, ACK] Seq=1 Ack=2 Win=65280 Len=0 TSval=3087585455 TSecr=3067521210
10…	10.014170572	88.198.46.51	192.168.1.93	TCP	66	34546 → 8096 [ACK] Seq=2 Ack=2 Win=7424 Len=0 TSval=3067521275 TSecr=3087585455

Checking conectivity domain1 to server with traceroute from OUTSIDE

1				*	*	*
2	core23.fsn1.hetzner.com	213.239.245.237	de	**3.969** ms	**3.949** ms	**3.930** ms
3	core0.fra.hetzner.com	213.239.224.82	de	**4.834** ms	**4.788** ms	**4.793** ms
4	static.185.2.47.78.clients.your-server.de	78.47.2.185	de	**19.643** ms	**19.625** ms	**19.639** ms
5	86-120-71-39.rdsnet.ro	86.120.71.39	ro	**34.980** ms	**34.961** ms	**34.939** ms
6				*	*	*
7				*	*	*
8				*	*	*
9				*	*	*
10				*	*	*

No reply for 5 hops. Assuming we reached firewall.

satitsu
9 marzo 2025 02:43

5

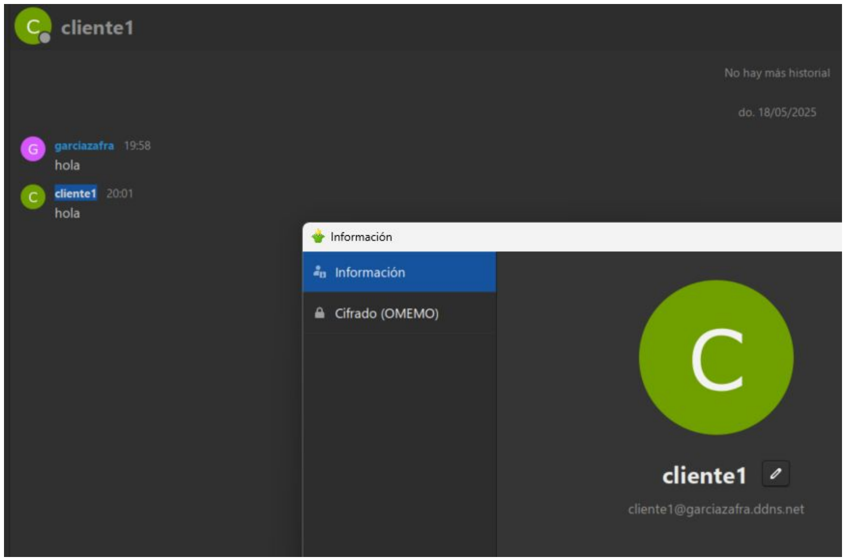
¿Has probado a reiniciar el router y que cambie de rango la IPv4? Hace un tiempo que ya comentamos por aquí que con las IPs que empiezan por 86.127.x.x no se puede acceder a la propia web de Digi. Es posible que tengan alguna cagada con el enrutamiento de esas IPs desde la propia red y por eso no puedas acceder. Verifica que ni la de tus padres ni la tuya estén en ese rango y vuelve a probar por si por casualidad es eso.

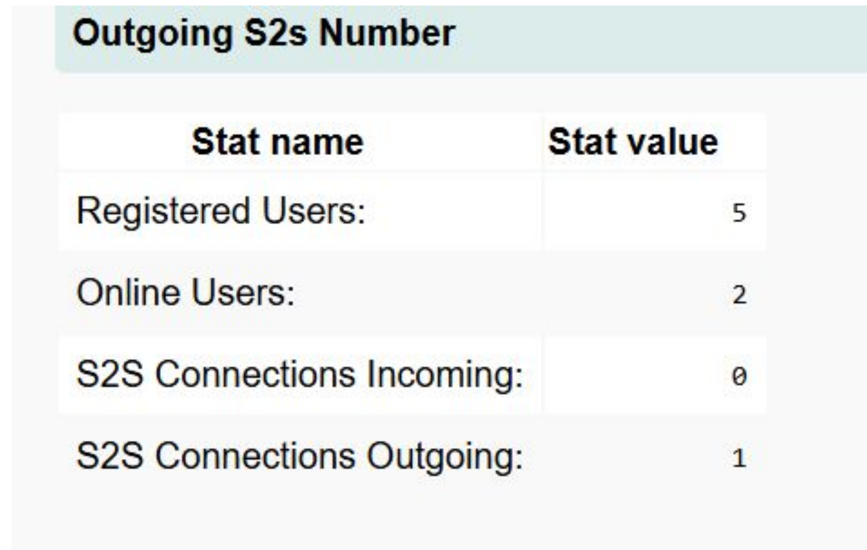
Saludos.

Checking conectivity domain2 to server with traceroute from OUTSIDE

```
C:\Users\Damian>tracert garciazafra.ddns.net

Traza a la dirección garciazafra.ddns.net [81.38.89.128]
sobre un máximo de 30 saltos:

  1     2 ms     1 ms     2 ms  10.183.192.1
  2     2 ms     3 ms     1 ms  static.masmovil.com [176.222.70.49]
  3    14 ms    13 ms    13 ms  10.200.254.205
  4    14 ms    14 ms    14 ms  10.97.1.30
  5    14 ms    14 ms    14 ms  10.97.1.29
  6    14 ms    14 ms    14 ms  10.15.2.26
  7    15 ms    16 ms    14 ms  212.231.61.219
  8     *        *        *     Tiempo de espera agotado para esta solicitud.
  9    27 ms    28 ms    27 ms  162.red-81-46-31.customer.static.ccgg.telefonica.net [81.46.31.162]
 10    26 ms    26 ms    26 ms  218.red-81-46-31.customer.static.ccgg.telefonica.net [81.46.31.218]
 11    27 ms    28 ms    27 ms  128.red-81-38-89.dynamicip.rima-tde.net [81.38.89.128]

Traza completa.
```

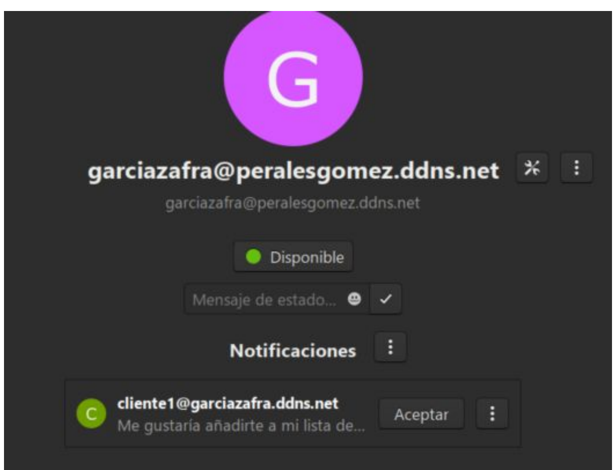
Show service 1



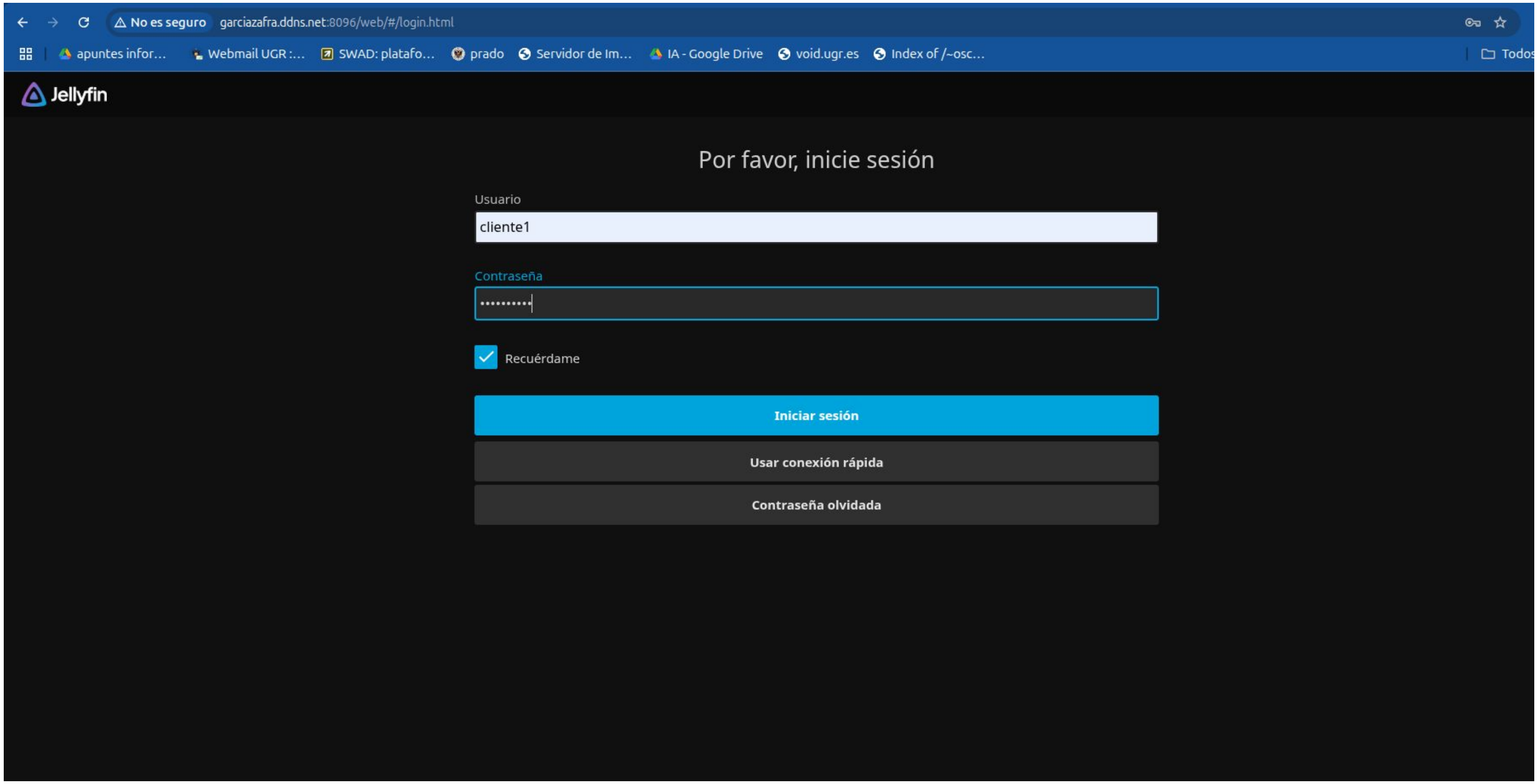


Outgoing S2s Number

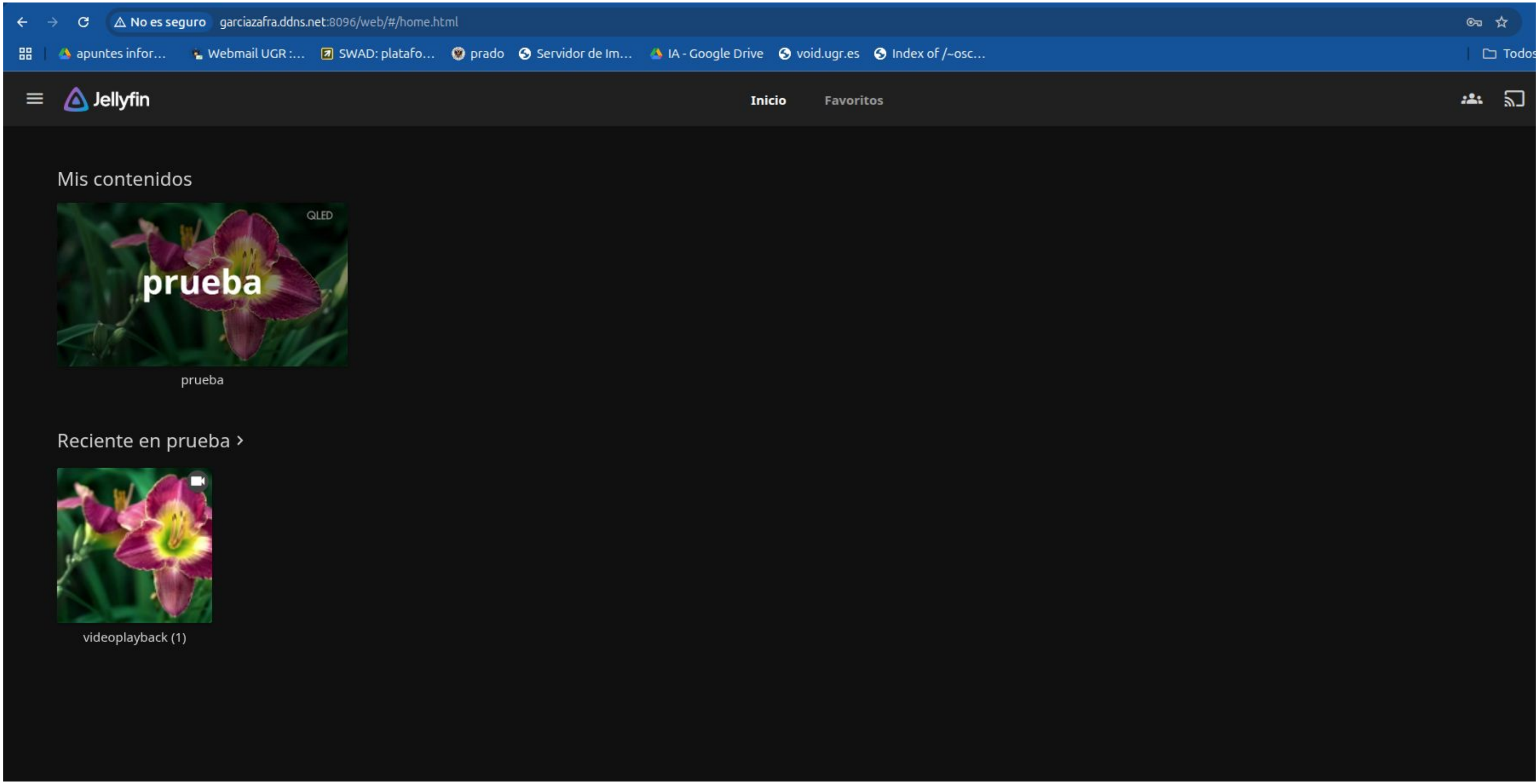
Stat name	Stat value
Registered Users:	5
Online Users:	2
S2S Connections Incoming:	0
S2S Connections Outgoing:	1



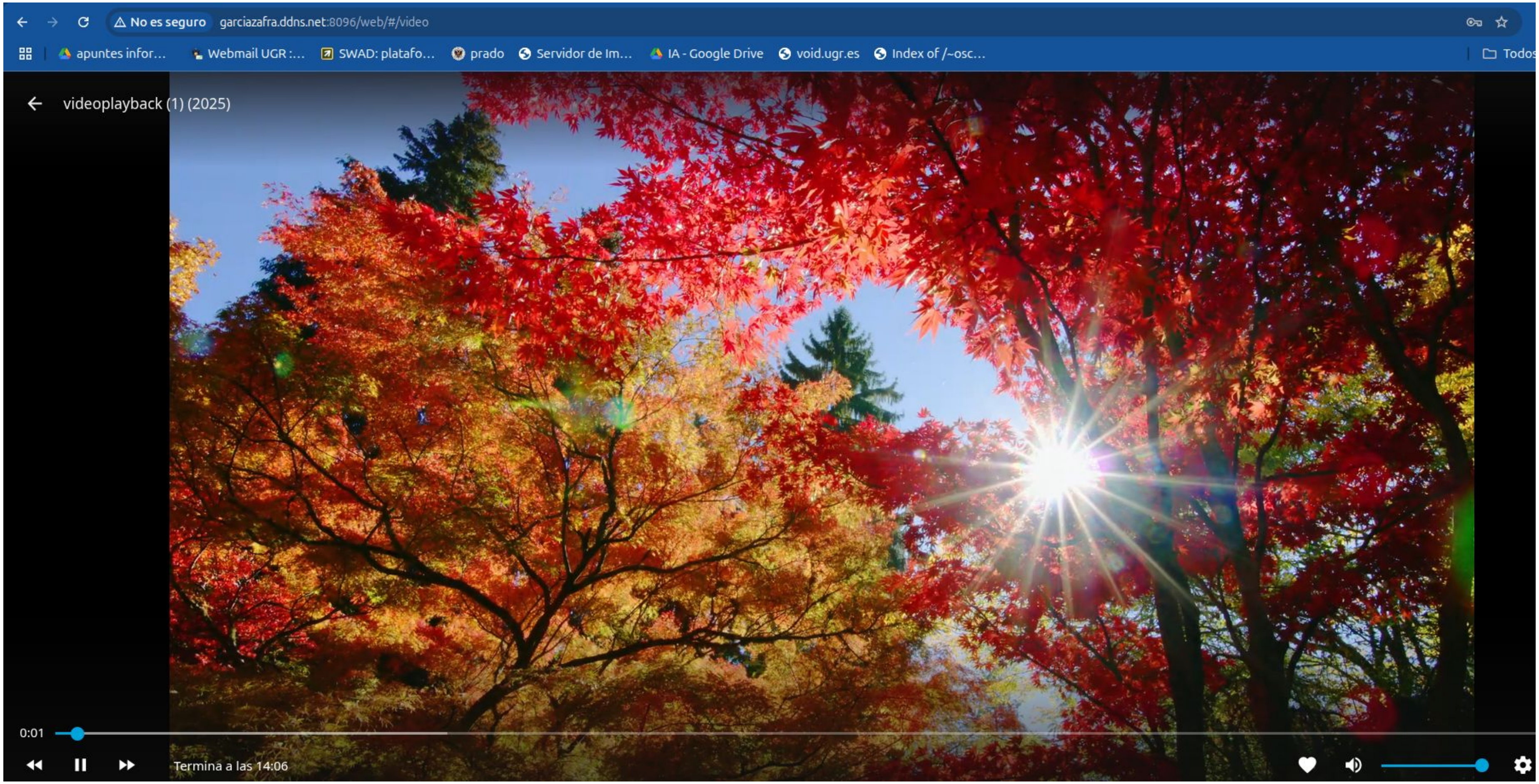
Show service 2



Show service 2



Show service 2



CONCLUSIONS

En esta práctica hemos aprendido a configurar servicios como XMPP y Jellyfin, lo que nos ha enseñado a enfrentarnos a diversos problemas que pueden presentarse al crear un servidor, como las complicaciones que implica estar detrás de una CG-NAT y cómo identificar esta situación. También hemos aprendido a abrir puertos y la importancia de configurar correctamente el firewall para que no bloquee las peticiones.

Sumado a ello, aprendimos a usar diversas herramientas de diagnóstico, como traceroute, ping u otras herramientas web, que nos permiten confirmar el correcto funcionamiento del servidor o identificar problemas en él. En general, esta práctica nos ha dado una experiencia muy útil para manejar problemas reales en el montaje de servicios en red, lo cual nos podrá ser útil en el futuro, ya sea si trabajamos en este campo o simplemente porque queramos montar un servidor por nuestra cuenta.

Por ejemplo, para descubrir si nos encontrábamos dentro de una cg-nat utilizamos traceroute para comprobar si se realizaban mucho saltos en direcciones privadas, además de comparar la dirección WAN de nuestro router con la proporcionada por whatismyip.com. Entramos en la configuración de los router para abrir los puertos 5222 (conexión cliente-servidor XMPP) y 5269 (conexión servidor-servidor XMPP), además del puerto 8096 para jellyfin.

También hemos configurado DNS dinámicos (DDNS), a través de la configuración del router, así como por el programa proporcionado por noip (DUC), con lo que la ip asociada al dominio se actualiza automaticamente.